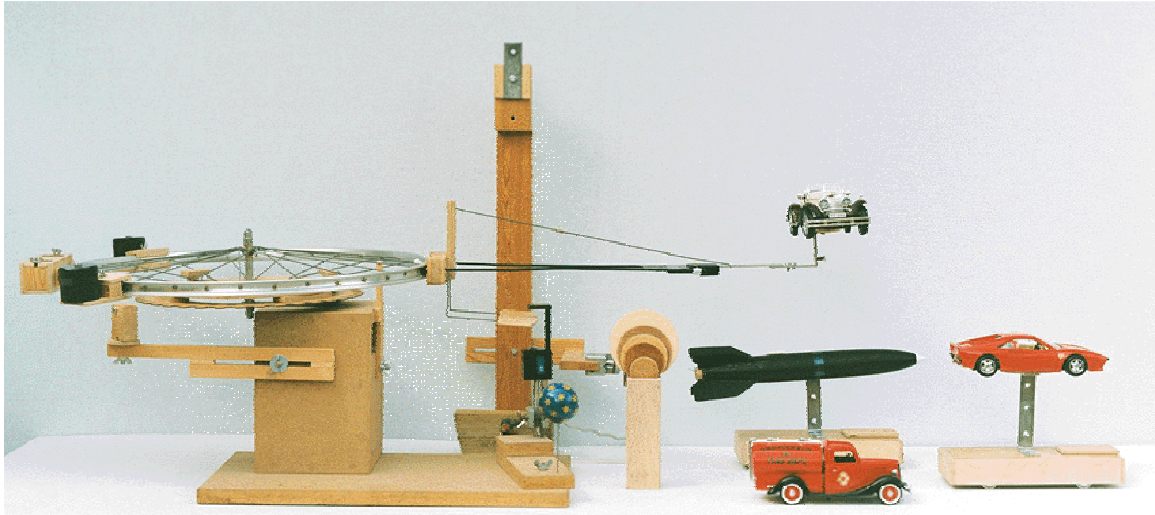


A merry-go-round as an alternative to the wind tunnel

by *Flemming Jørgensen*



Outline

- 1: Introduction
- 2: Air resistance and the drag coefficient C_d
- 3: The circular motion of a model mounted on the merry-go-round
- 4: How to measure air resistance and C_d by considering the model point as being fixed to a car moving along a linear s-axis.
- 5: Measuring the model point mass.
- 6: Results for a sphere, a circular disk and a vintage car
- 7: The drag coefficient for some particularly chosen car models
- 8: Reynolds model law
- 9: A model rocket

Appendices

- A: Solving the differential equations
- B: The equivalent car and its mass
- C: Use of a spreadsheet
- D: A good starting point for the fitting procedure
- E: Determination of the frontal area
- F: Table of drag coefficients
- R: References

